Rachel Lin

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EDUCATION/AWARDS

Amherst College | Amherst, MA **May 2025**

- Bachelor of Arts: Computer Science **GPA**: 3.74/4
- Relevant Coursework: Computer Systems, Statistics via Modeling, Linear Algebra via Applications, Data Structures, Algorithms, Artificial Intelligence, Data Science, Machine Learning, Data Mining

Major League Hacking CivicHacks Hackathon | *Third Place, Best Accessibility Hack* **January 2023**

CodePath Technical Program | Completed *Applications of AI Engineering* with *Honors* **|** AI Open Source Capstone **Sept 2026**

EXPERIENCE

Full Stack Develop er @ IHAI Time (Initiative) | Remote **July 2025 - Present**

- Developing a volunteer management platform using React, FastAPI, and PostgreSQL to support shift coordination, approvals, and time-credit tracking.
- Designed backend services, REST APIs, and relational data models; developing AI-assisted spreadsheet imports to standardize volunteer data and reduce manual data entry; evaluated on synthetic datasets, reducing 100-row import time by ~97% with 98.1% field-level accuracy

Full Stack Developer (Volunteer) @ Keelung For a Walk Program | Remote | *Keelung, Taiwan* **February 2026 - Present**

- Improving admin and user experiences in a Vue.js/Typescript SPA by building admin editor tools through multi-code structure support and user experience through WebAR challenges, location-based interactions, completion pages with timer functionality, and multilingual map synchronization, supporting delivery of 200+ unique challenge codes

Data Mining Intern | Amherst College | Amherst, MA **January 2025 - May 2025**

- Built a tweet data pipeline using Selenium to collect and analyze 10,000+ tweets; used BETO (*Spanish BERT*) to generate sentiment labels and compared Logistic Regression and Random Forest classifiers to analyze trends across Latin American feminist hashtags ; Developed 15+ visualizations with *pandas* + *seaborn* to communicate engagement and temporal trends across hashtags.

COSC-112 Teaching Assistant | Amherst College | Amherst, MA **September 2023 - December 2023**

- Led weekly lab sessions and 12+ office hours for ~40 students in Intro to Computer Science II, supporting projects and coursework in core artificial intelligence concepts and data structures and algorithms

Research Fellowship → Gregory Call Intern | Amherst College | Amherst, MA **June 2023 - December 2023**

- Built a Python discrete-event simulation of Starbucks ordering across three channels (mobile, drive-thru, in-person), modeling 25 million customer entries across 25 queue-threshold scenarios
- Analyzed customer switching and balking behavior using *NumPy* and *pandas*; visualized response-time and customer-retention tradeoffs with *Matplotlib* and *Seaborn*

PROJECT

FundFlow — Fund Holdings Pipeline | *Python, Playwright, AWS S3, MongoDB Atlas, FastAPI, pytest* **October 2026**

- Built an end-to-end data pipeline processing 19,000+ holdings across 5 iShares ETFs, preserving source reports in AWS S3, verifying SHA-256 checksums and row-level validation before loading data into MongoDB Atlas.
- Orchestrated ingestion through a manually triggered GitHub Actions workflow with AWS OIDC authentication, automated tests, and run summaries; implemented batch upserts, duplicate prevention, and source traceability, with a deployed FastAPI dashboard for inspecting holdings and processing outcomes.

U.S Hurricane Landfall Detector | *Python, GeoPandas, Shapely, PyProj, pytest, Streamlit* **August 2026**

- Built an end-to-end Python geospatial analytics pipeline that parses NOAA HURDAT2 storm records, reconstructs via geometric interpolation, and detects 488 hurricane-strength landfall candidates across 18 Atlantic and Gulf coastal states since 1900 with *Shapely/GeoPandas*, and interpolates landfall time and wind speed for analysis and reporting.
- 94.9% reference coverage across all matching candidates; implemented 56 automated tests and GitHub Actions workflows for validated data updates and dashboard availability checks, with automatic deployment to Streamlit.

OTHER/ACTIVITIES

The Remedy Project (Civic Advocacy Organization) | Remote **January 2023 - February 2024**

Lead Data & IT Team Member (Volunteer)

- Managed backend operations of the action-page platform using NationBuilder, supporting supporter data workflows, CRM segmentation, donation pipelines, and campaign automation for 1,000+ community members
- Collaborated with co-founders to improve website architecture and user experience by customizing templates with Jinja, HTML/CSS, and JavaScript, building dynamic landing pages and streamlining action-page conversions

SKILLS & INTERESTS

Languages: Python, SQL, Java, JavaScript/TypeScript, R
Data & Databases: PostgreSQL, MongoDB Atlas, pandas, NumPy, ETL, NLP
Infrastructure & Testing: AWS S3, GitHub Actions/CI, Docker, Git, pytest, RAG
Frameworks & Collection: FastAPI, Selenium, Playwright, React, Vue.js